

LA BÓVEDA

"Trust No Cloud."

Offline-First Local Security Vault
User & System Reference Manual

Software Version: 1.0.0 (Premium Release)

Architecture: Offline, Client-Side Only

Attribution: Made by Origin Labs

Published Date: June 28, 2026

1. Introduction & Security Paradigm

La Bóveda is a premium, client-side offline-first personal security vault designed to store credentials, notes, and file attachments without relying on remote servers, cloud synchronization, or network telemetry. Built upon the philosophy of 'Trust No Cloud', the application operates inside a completely local sandbox environment in the user's web browser.

Security Architecture & Cryptographic Foundations

The vault employs industry-standard client-side security mechanics to ensure that your sensitive information remains secure, private, and tamper-proof. While avoiding the specifics of standard algorithms, the vault functions through the following three security pillars:

1. Master Password & Key Derivation: The user sets a Master PIN, which is not stored directly on the device.

Instead, the application runs the password through a heavy key-

derivation function using a cryptographically secure salt. By applying several hundred thousand rounds of iterations, it derives a strong secondary cryptographic key. This high-computational process prevents fast dictionary or brute-force attacks on the Derived Key.

2. Symmetric Data Encryption: Every vault item, including credentials, notes, and uploaded files, is encrypted individually. The app generates an initialization vector (IV) for each entry and encrypts the payload using a modern, authenticated symmetric block cipher (employing 256-bit keys). This block cipher mode ensures not only absolute data confidentiality but also integrity verification, rendering unauthorized tampering easily detectable during decryption.

3. Local Database & Sandboxing: The database is stored completely offline inside the user's browser, utilizing native local storage tables. The system divides files and metadata stores to ensure high performance. No telemetry, analytical beacons, or cloud backdoors are compiled into the codebase, guaranteeing total single-device sovereignty.

2. Core Vault Capabilities

Credentials Vault

The Credentials module enables users to securely store accounts, passwords, and related metadata. Key aspects include:

- **Custom Categories:** Sort credentials into login, financial, personal, or work folders.
- **Live Strength Analyzer:** Evaluates the complexity and entropy of passwords in real time.
- **Clipboard Auto-Clear:** Quickly copy user attributes or passwords to the clipboard, with a built-in safeguard that automatically purges copied data from the clipboard after a secure duration.

- **Quick Filters:** Tag and star frequently used credentials for instant access.

Secure Notes

The Secure Notes module is a local text container for keeping sensitive personal information, server configurations, recovery codes, or private correspondence. Notes are categorized, fully searchable, and stored under the derived symmetric master key. Features include custom tagging, favorite triggers, and responsive search indexing.

Secure Offline File Vault

Unlike standard password managers, La Bóveda supports secure offline file attachments. Files are read locally via the browser, converted to binary arrays, and encrypted directly using the AES key. The encrypted binary data is stored in a dedicated local database sub-table to maintain fast index scans on standard credential entries. Users can securely download files back to their local machine and perform secure deletion to clear them from storage.

Custom Entropy Generator

The Entropy Generator generates highly secure passwords on the fly. Users can customize options such as:

- Length adjustment (from basic to highly secure combinations).
- Inclusion of uppercase letters, lowercase letters, numbers, and specialized symbols.
- Generation of memorable, readable passphrases containing specific custom separators.

Operations Center (Dashboard)

The Operations Center serves as the primary command center. It provides an immediate overview of the vault's status, including total password count, notes, files, last login timestamp, and recent activity logs. It also hosts the

'Panic' button for instant vault locking and features shortcuts for rapid item creation.

3. Stealth Camouflage System

To protect against physical coercion or shoulder-surfing, La Bóveda features a stealth decoy system:

1. **Weather Camouflage Screen:** When activated, the application displays a fully functional weather interface showing weather statistics (simulated or loaded from standard public telemetry). To a casual onlooker, the application appears to be a basic utility dashboard.
2. **Hidden Gesture Unlock:** The entry point to the actual PIN prompt is hidden behind configurable gestures, such as double tapping the temperature reading or holding down the city title for a specific duration. Only when these gestures are correctly performed will the interface transition to the secret login screen.

4. Advanced Operator Deck (Black Room)

Accessible through an easter egg trigger, the Black Room acts as an offline cryptotool suite. It operates locally inside the browser memory without network calls. Tools include:

- **Text Hasher:** Calculates secure cryptographic digests of text strings.
- **Base64 Encoder/Decoder:** Encodes and decodes base64 and URL-encoded strings.
- **File Checksum Tool:** Computes local hash values of files to verify integrity.
- **Password Estimator:** Analyzes passwords to calculate entropy bits and estimated crack duration.
- **JSON Linter/Formatter:** Formats, validates, and minifies JSON inputs locally.
- **QR Code Creator:** Generates offline QR codes directly on the canvas.
- **Secure Random Generator:** Generates cryptographically

secure random bytes in hex, base64, or integer sequences.

5. System Operations & Data Portability

The Control Room provides overall management features:

- **Import/Export Backup:** Users can download a JSON-based encrypted backup of their entire database. This backup can be imported into another device running the vault locally, ensuring complete data portability without cloud synchronization.
- **Database Wipe:** An absolute reset function ('ELIMINAR TODO') immediately purges all database tables from the browser's storage, wiping all credentials, salts, and session caches instantly.